

QR Generator Project Report (Python)

This project demonstrates how to generate QR (Quick Response) codes using Python. A QR code is a two-dimensional barcode that stores information such as website URLs, text, contact details, and more. Users can scan the generated QR code using a mobile device to access the stored information instantly.

Project Objectives

- Generate QR codes from URLs.
- Save QR codes as image files.
- Customize QR code appearance using colors and settings.
- Demonstrate the use of Python libraries such as qrcode and Pillow (PIL).

Technologies Used

- Python
- qrcode Library
- Pillow (PIL) Library
- VS Code / Any Python IDE

Code 1: Basic QR Generator

The first program creates a simple QR code from a YouTube link and saves it as an image. It is the easiest way to generate a QR code using the qrcode library.

Working: 1. Import qrcode library. 2. Use qr.make() to create a QR code. 3. Pass the URL to be encoded. 4. Save the generated image using save().

Code 2: Advanced QR Generator

The second program provides more control over the QR code. It allows customization of error correction, size, border, and colors.

Parameters Explained:

- version = 1 : Defines QR code size.
- ERROR_CORRECT_H : High error correction level.
- box_size = 10 : Size of each QR box.
- border = 4 : Thickness of the border.
- fill_color = "red" : QR code color.
- back_color = "white" : Background color.

Advantages

- Fast and easy information sharing.
- Contactless access to websites and resources.
- Easy customization.
- Can be used in marketing, education, and business applications.

Applications

- Website links
- Payment systems
- Product information
- Attendance systems
- Event registration
- Digital business cards

Conclusion

The QR Generator project is a beginner-friendly Python application that demonstrates the use of external libraries, image generation, and data encoding. The project can be further enhanced by creating a graphical user interface, adding logo support, and allowing users to generate QR codes for different types of data.